

## Multiple Choice

1. **Answer:** C

*Options:* A) Lipids are the primary source of energy for cellular processes.; B) Lipids are soluble in water but insoluble in organic solvents.; C) Phospholipids are water soluble.; D) Steroids are lipids that consist of glycerol and fatty acids.; E) Lipids make up most cell surface receptors.

*Note:* Correct option: C - Phospholipids are water soluble.

2. **Answer:** A

*Options:* A) a universally accepted fact about the origin and development of all species.; B) a hypothesis in the process of being tested and verified.; C) a disproven theory about the development of species over time.; D) a speculation about possible changes in populations over time.; E) an opinion that some scientists hold about how living things change over time.

*Note:* Correct option: A - a universally accepted fact about the origin and development of all species.

3. **Answer:** D

*Options:* A) By observing the cell under a microscope; B) Using fluorescent tagging of proteins; C) Tracking changes in cell size and shape; D) Autoradiography with labelled amino acids; E) Through protein synthesis

*Note:* Correct option: D - Autoradiography with labelled amino acids

4. **Answer:** A

*Options:* A) Plants, animals, and fungi all have both chloroplasts and mitochondria.; B) Plants and fungi have chloroplasts but no mitochondria; animals have only mitochondria.; C) Animals have both chloroplasts and mitochondria; plants and fungi have only chloroplasts.; D) Plants and animals have chloroplasts but no mitochondria; fungi have only mitochondria.; E) Plants and animals have mitochondria but no chloroplasts; fungi have only chloroplasts.

*Note:* Correct option: A - Plants, animals, and fungi all have both chloroplasts and mitochondria.

5. **Answer:** C

*Options:* A) Appetitive behavior, exploratory behavior, quiescence; B) Termination, appetitive behavior, exploratory behavior; C) Appetitive behavior, consummatory act, quiescence; D) Consummatory act, exploratory behavior, termination; E) Appetitive behavior, termination, consummatory act

*Note:* Correct option: C - Appetitive behavior, consummatory act, quiescence

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'By using operant conditioning'.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'tracheids and vessel elements'.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'stabilizing selection'.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'polyspermy reaction'.

## Long Form Response

11. **Answer:** Courtship serves as a means of communication, sexual identification, species identification, and in some cases, a necessary condition for egg production.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Living things are characterized by a specific, complex organization, the ability to consume energy, movement, response to physical or chemical changes, growth, reproduction, and adaptability.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*